

Department of Physical Science
Faculty of Applied Science
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Parallel Computing - IT3243

Tutorial No 03: Interconnection Networks for Parallel Computers

1. List out the factors that impact the performance of a serial or implicitly parallel program. How it is resolved.
2. Describe the typical SIMD architecture and a typical MIMD architecture.
3. Explain two forms/platform of data exchange model between parallel tasks.
4. Why define NUMA and UMA architectures only in terms of memory access times and not cache access times.
5. Describe the ideal parallel model, PRAM (Parallel Random-Access Machine) model.
6. List out four subclasses of PRAM base on how simultaneous memory accesses are handled.
7. List the protocols that are used to resolve concurrent writes in PRAM.
8. Discuss the following network topologies:

Bus Crossbar Multistage Networks Completely connected
network

Linear array Mesh Hypercube Tree

9. Discuss various criteria used to characterize the cost and performance of static-interconnection networks.
 - a. Diameter
 - b. Connectivity
 - c. Bisection Width
 - d. Cost
10. Determine the following characteristics of cost and performance of each of the interconnection network and fill it out.
11. Define the following times relates to transfer a message over a network:
 - a. Startup time (t_s):
 - b. Per-hop time (t_h):
 - c. Per-word transfer time (t_w):

12. Describe what is meant by store-and-forward routing and cut-through routing in message passing with the aid of an example.

Network	Diameter	Bisection Width	Arc Connectivity	Cost (No. of links)
Completely-connected				
Star				
Complete binary tree				
Linear array				
2-D mesh, no wraparound				
2-D wraparound mesh				
Hypercube				
Wraparound k -ary d -cube				

Network	Diameter	Bisection Width	Arc Connectivity	Cost (No. of links)
Crossbar				
Omega Network				
Dynamic Tree				